

● HARD

● GEOMETRY AND TRIGONOMETRY

Geometry and Trigonometry

13

4 questions · ~6 min

Q1

GEOMETRY AND TRIGONOMETRY

$$(x-6)^2 + (y+5)^2 = 16$$

In the xy -plane, the graph of the equation above is a circle. Point P is on the circle and has coordinates $(10, -5)$. If $\overline{PQ}$ is a diameter of the circle, what are the coordinates of point Q ?

- A. $(2, -5)$
- B. $(6, -1)$
- C. $(6, -5)$
- D. $(6, -9)$

An isosceles right triangle has a hypotenuse of length 58 inches. What is the perimeter, in inches, of this triangle?

- A. $29\sqrt{2}$
- B. $58\sqrt{2}$
- C. $58 + 58\sqrt{2}$
- D. $58 + 116\sqrt{2}$

The equation $x^2 + y - 1^2 = 49$ represents circle A. Circle B is obtained by shifting circle A down 2 units in the xy -plane. Which of the following equations represents circle B?

A. $x - 2^2 + y - 1^2 = 49$

B. $x^2 + y - 3^2 = 49$

C. $x + 2^2 + y - 1^2 = 49$

D. $x^2 + y + 1^2 = 49$

Q4

GRID-IN

GEOMETRY AND TRIGONOMETRY

Point O is the center of a circle. The measure of arc RS on this circle is 100° . What is the measure, in degrees, of its associated angle ROS ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.